

Doob's Martingale Convergence Theorem in Mathlib

A walk through Ying & Degenne's CPP'23 formalisation

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Plan

The topics will be covered in this talk are:

- ▶ Mathematical background and corresponding design of formalization on martingales and the convergence theorem;
- ▶ The proof sketch of martingale convergence theorem.

Here's the general setting of the formalization:

```
variable {Ω ι E k : Type*} [RCLike k] {G F : MeasurableSpace Ω}
  {μ : Measure Ω} {f g : Ω → E} {s : Set Ω} [Preorder ι]
```

Here $(\Omega, \mathcal{F}, \mu)$ is a measure space, $\mathcal{G} \subset \mathcal{F}$ is a sub- σ -algebra, E often refers to a Banach space (e.g. $\mathbb{R}$), and ι refers to an index set.

Note: this paper is implemented in Lean 3 but I've updated the codes in this paper with the current codes in Lean 4.

Outline

Background on Measure theory and Probability theory

Stochastic Processes and Martingales

Stopping Times and Upcrossings

Doob's upcrossing estimate

Convergence Theorems

Revised version of Stopping Times in current Mathlib

Measurable spaces and measures

Let $(\Omega, \mathfrak{F}, \mu)$ be a measure space, which consists of a type Ω , a σ -algebra $\mathfrak{F}$, and μ be a measure on $(\Omega, \mathfrak{F})$.

Definition

A measure μ is said to be **finite** if $\mu(\Omega) < \infty$, and is called **σ -finite** if there is an increasing sequence of measurable sets $(S_n)_{n \in \mathbb{N}}$ with finite measure s.t. $\bigcup_{n \in \mathbb{N}} S_n = \Omega$. If $\mu(\Omega) = 1$, then μ is said to be a probability measure.

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Remark

Although this talk is about objects like stochastic processes, which are often within the context of probability theory, none of the results that we will mention later will require μ to be probability measure. Most of the results will not use any assumption on the measure, while the others might require it to be σ -finite or finite.

Different notions of measurability in Lean

In mathlib, it contains several notions of measurability of functions. Apart from commonly-used 'measurable', we will introduce two notions that are most relevant to this talk: `StronglyMeasurable` and `AEStronglyMeasurable`.

Definition

A **simple function** $f : \Omega \rightarrow E$ is a function whose every preimage $f^{-1}(\{x\})$ is measurable and whose range is finite.

```
structure SimpleFunc ( $\Omega$  : Type*) [MeasurableSpace  $\Omega$ ] ( $E$  : Type*) where  
  toFun :  $\Omega \rightarrow E$   
  measurableSet_fiber' :  $\forall x$ , MeasurableSet (toFun  $^{-1}$ , {x})  
  finite_range' : (Set.range toFun).Finite
```

Definition

A function $f : \Omega \rightarrow E$ is called **strongly measurable** if it is the pointwise limit of a sequence of simple functions.

A function $f : \Omega \rightarrow E$ is called **a.e. strongly measurable** w.r.t. μ if there exists a strongly measurable g such that $f = g$ μ -a.e.

```
def StronglyMeasurable (f :  $\Omega \rightarrow E$ ) : Prop :=  
   $\exists$  fs :  $\mathbb{N} \rightarrow \Omega \rightarrow_s E$ ,  $\forall \omega$ , Tendsto (fun n => fs n  $\omega$ ) atTop ( $\mathcal{N}$  (f  $\omega$ ))
```

```
def AEStronglyMeasurable (f :  $\Omega \rightarrow E$ ) ( $\mu$  : Measure  $\Omega$ ) : Prop :=  
   $\exists$  g :  $\Omega \rightarrow E$ , StronglyMeasurable g  $\wedge$  f  $=^m[\mu]$  g
```

Recall that one can easily define integral on simple function. For any function $f : \Omega \rightarrow \mathbb{R}_{\geq 0} \cup \{\infty\}$, we can define the **Lebesgue integral** of f as the supremum of the integrals of all (pointwise) smaller simple functions. This is denoted $\int_{\Omega} f \, d\mu$.

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- ▶ a strongly measurable function can be approximated by simple functions;
- ▶ the Bochner integral is the limit of the integrals of such simple functions.

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So in this talk, in order to use Bochner integral, we will ask functions to be strongly-measurable/ a.e. strongly-measurable rather than measurable.

So in this talk, in order to use Bochner integral, we will ask functions to be strongly-measurable/ a.e. strongly-measurable rather than measurable. However, in 2nd-countable metrizable spaces with Borel σ -algebra, 'strongly-measurable' and 'measurable' are equivalent. So in most of the cases (e.g. $\mathbb{R}^d$) one can safely ignore the difference.

Random variables and Expectation

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Definition

We denote by $L^0(\Omega, E; \mu)$ the space of **a.e.strongly-measurable** functions, quotiented by a.e. equal relation. For $p \geq 1$, we define $L^p(\Omega, E; \mu)$ as follow:

$$L^p(\Omega, E; \mu) = \left\{ f \in L^0(\Omega, E; \mu) : \int_{\Omega} \|f(x)\|^p d\mu(x) < \infty \right\}.$$

The functions in $L^1(\Omega, E; \mu)$ are said to be **integrable**. On this L^1 space, the Bochner integral is well-defined and we can introduce $\mu[f] = \int_{\Omega} f(x) d\mu(x)$, which is called **expectation** of f if μ is a probability measure.

Conditional Expectation

In contrast to expectation, which can be viewed as the best forecast of a random variable without extra information, we can view **conditional expectation** as the best forecast of a random variable when we are given partial information $\mathcal{G} \subset \mathcal{F}$.

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Definition

Let $(\Omega, \mathfrak{F}, \mu)$ be a measure space and let $\mathfrak{G}$ be a sub- σ -algebra of $\mathfrak{F}$. Let E be a Banach space and $f : \Omega \rightarrow E$ be an integrable function w.r.t. μ . Then a function $g : \Omega \rightarrow E$ is said to be a **conditional expectation** of f w.r.t. $\mathfrak{G}$ if it is integrable, $\mathfrak{G}$ -(strongly) measurable and for all $\mathfrak{G}$ -measurable sets s ,

$$\int_{x \in s} g(x) d\mu(x) = \int_{x \in s} f(x) d\mu(x)$$

If the measure μ restricted to the σ -algebra $\mathfrak{G}$ is σ -finite, then there exists a conditional expectation and it is unique up to a.e. equality. We denote it as $\mu[f|\mathfrak{G}]$.

Definition

A **stochastic process** $(f_i)_{i \in \iota}$ on Ω is an ι -indexed family of random variables on Ω , where the index set is ordered and usually represents time. Most often, $\iota = \mathbb{N}$ or $\iota = \mathbb{R}_{\geq 0}$.

Stochastic process, filtration and martingales

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Definition

A **filtration** $(\mathfrak{F}_i)_{i \in \iota}$ of $\mathfrak{F}$ is a non-decreasing sequence of sub- σ -algebras of $\mathfrak{F}$. This often represents information available up to some time. We call $(\Omega, \mathfrak{F}, \mu, (\mathfrak{F}_i)_i)$ a **filtered space**.

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Definition

A stochastic process $(f_i)_{i \in \iota}$ is said to be **adapted** to a filtration $(\mathfrak{F}_i)_{i \in \iota}$ if for all $i \in \iota$, f_i is $\mathfrak{F}_i$ -strongly measurable.

In Lean, we write a stochastic process as $f : \iota \rightarrow \Omega \rightarrow E$ with $[\text{Preorder } \iota]$. For filtration, it is formalized as follow:

```
structure Filtration { $\Omega$  : Type*} ( $\iota$  : Type*) [ $\text{Preorder } \iota$ ] ( $\mathfrak{F}$  :  
  MeasurableSpace  $\Omega$ ) where  
  seq :  $\iota \rightarrow \text{MeasurableSpace } \Omega$   
  mono' : Monotone seq  
  le' :  $\forall i : \iota, \text{seq } i \leq \mathfrak{F}$ 
```

For the notion of 'adapted', it is formalized as follow:

```
def StronglyAdapted ( $\mathcal{F}$  : Filtration  $\iota$   $\mathfrak{F}$ ) ( $f : \iota \rightarrow \Omega \rightarrow E$ ) : Prop :=  
   $\forall i : \iota, \text{StronglyMeasurable}[\mathcal{F} \ i] \ (f \ i)$ 
```

Note: recently Adapted has been renamed as StronglyAdapted, and Adapted has been reserved for the case when f_i are $\mathfrak{F}_i$ -measurable rather than strongly-measurable.

Example of stochastic process

Example

At time $t \in \mathbb{N}$, a coin is flipped and the player gains 1 if the result of the flip is heads, and loses 1 if it is tails. The amount of money of the player over time is a stochastic process $(f_n)_{n \in \mathbb{N}}$, which takes values in $\mathbb{R}$.

Martingale

Definition

Let $(\Omega, \mathfrak{F}, \mu, (\mathfrak{F}_i)_{i \in \mathbb{N}})$ be a filtered space. A stochastic process $(f_i)_{i \in \mathbb{N}}$ is called **martingale** if

- ▶ It is adapted to the filtration $(\mathfrak{F}_i)_{i \in \mathbb{N}}$;
- ▶ f_i is integrable for all i ;
- ▶ the conditional expectation $\mu[f_i | \mathfrak{F}_{i-1}] = f_{i-1}$ for all i .

We call $\forall i, \mu[f_i | \mathfrak{F}_{i-1}] = f_{i-1}$ the martingale property. It is equivalent to say: for all i, j with $i \leq j$, $\mu[f_j | \mathfrak{F}_i] = f_i$, as one can check.

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Remark

*The martingale property tells us that given the information up to time $i - 1$, the best forecast of the value of f_i is just the value f_{i-1} at the moment. This suggests that **martingale models a series of fair games**, as the player is not expected to gain/lose based on the information over the past time.*

Example

Recall the example of coin flip, which is an example of series of fair games. Let X_i be the random variable describing the coin flip at time i and let $\sigma(X_i)$ be the σ -algebra generated by X_i . Set $\mathfrak{F}_i := \sigma(X_j : j \leq i)$ and let $f_i = \sum_{j \leq i} X_j$ be random variable describing the amount of money of the player at time i . One can see that:

- ▶ (f_i) is adapted to $(\mathfrak{F}_i)$;
- ▶ f_i is integrable for all i ;
- ▶ $\mu[f_i | \mathfrak{F}_{i-1}] - f_{i-1} = \mu[f_i - f_{i-1} | \mathfrak{F}_{i-1}] = \mu[X_i | \mathfrak{F}_{i-1}] = \mu[X_i] = 0$.

So this is indeed an example of martingale. This is called an example of **simple random walk**.

In Lean, we consider index set to be ordered ι as well:

```
def Martingale (f :  $\iota \rightarrow \Omega \rightarrow E$ ) (F : Filtration  $\iota$   $\mathfrak{F}$ ) ( $\mu$  : Measure  $\Omega$ ) :
Prop := StronglyAdapted F f  $\wedge \forall i j, i \leq j \rightarrow \mu[f_j | \mathcal{F}_i] =^m [\mu] f_i$ 
```

Note we don't ask f_i to be integrable in Lean's definition, but as we put junk value 0 for $\mu[f_j|\mathfrak{F}_i]$ when f_j is not integrable, assume f_j is not integrable, then for all $i \leq j$,

$$f_i = \mu[f_j|\mathfrak{F}_i] = 0 \text{ a.e.}$$

which in particular forces $f_j = 0$ a.e., which is not possible as 0 is integrable.

Submartingale and supermartingale

Martingale gives a model for a series of fair games, while for biased games, we have the notion of **submartingale** and **supermartingale**:

Definition

Let $(\Omega, \mathfrak{F}, \mu, (\mathfrak{F}_i)_{i \in \mathbb{N}})$ be a filtered space. Given a stochastic process $(f_i)_{i \in \mathbb{N}}$ that is

- ▶ adapted to the filtration $(\mathfrak{F}_i)_{i \in \mathbb{N}}$;
- ▶ f_i is integrable for all i .

If $\mu[f_i | \mathfrak{F}_{i-1}] \leq f_{i-1}$, then we call it **supermartingale**; if $\mu[f_i | \mathfrak{F}_{i-1}] \geq f_{i-1}$, then we call it **submartingale**.

In Lean, we define submartingale and supermartingale as follow:

```
def Supermartingale [LE E] (f :  $\iota \rightarrow \Omega \rightarrow E$ ) ( $\mathcal{F}$  : Filtration  $\iota$   $\mathfrak{F}$ ) ( $\mu$  :  
  Measure  $\Omega$ ) : Prop :=  
  StronglyAdapted  $\mathcal{F}$  f  $\wedge$  ( $\forall$  i j, i  $\leq$  j  $\rightarrow$   $\mu[f$  j |  $\mathcal{F}$  i]  $\leq^m[\mu]$  f i)  $\wedge$   $\forall$  i,  
    Integrable (f i)  $\mu$   
  
def Submartingale [LE E] (f :  $\iota \rightarrow \Omega \rightarrow E$ ) ( $\mathcal{F}$  : Filtration  $\iota$   $\mathfrak{F}$ ) ( $\mu$  :  
  Measure  $\Omega$ ) : Prop :=  
  StronglyAdapted  $\mathcal{F}$  f  $\wedge$  ( $\forall$  i j, i  $\leq$  j  $\rightarrow$  f i  $\leq^m[\mu]$   $\mu[f$  j |  $\mathcal{F}$  i])  $\wedge$   $\forall$  i,  
    Integrable (f i)  $\mu$ 
```

In contrast to martingale, we do ask f_i to be integrable here.

No-system theorem for martingale

As martingale models a series of fair games, one should not be expected to systematically gain benefits from such games. This is called 'no-system theorem for martingale'. One can formulate this theorem as follow:

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Definition

A stochastic process $(c_n)_{n \in \mathbb{N}}$ is said to be **predictable** w.r.t the filtration $(\mathfrak{F}_n)_{n \in \mathbb{N}}$ if for all n , c_{n+1} is $\mathfrak{F}_n$ -measurable.

Theorem

If (f_n) is a (sub/super)-martingale adapted to the filtration $(\mathfrak{F}_n)$ and (c_n) is predictable, non-negative, and bounded, then

$$(c \cdot f)_n := \sum_{k=0}^{n-1} c_{k+1}(f_{k+1} - f_k)$$

is also a (sub/super)-martingale. For supermartingale case, we call it 'no-system theorem'.

One can think of this theorem in the following concrete case: one can think of stock price as an example of (sub/super)-martingale (under certain conditions). Naively, we can view the stock price of an intrinsically growing business as submartingale, and stock price of an intrinsically declining business as supermartingale.

Then one can think of:

- ▶ c_n as the number of shares an investor decides to hold at time $n - 1$ up to time n ;
- ▶ $(c \cdot f)_n = \sum_{k=0}^{n-1} c_{k+1}(f_{k+1} - f_k)$ represents all the wealth the investor accumulated up to time n .

'No-system theorem' in some sense tells that (under certain condition) one should not expect to systematically gain money from the falling stock. This notion will be helpful for later understanding 'upcrossing'.

Stopping time

In stochastic process, we often want to consider random times that are determined by occurrence of a particular event. If this event depends only on what happens in the past, we call it **stopping time**:

Definition

Given a ι -indexed filtration $(\mathfrak{F}_i)_{i \in \iota}$, we say a function $\tau : \Omega \rightarrow \iota$ is a **stopping time** if for all $i \in \iota$, $\{\tau \leq i\}$ is $\mathfrak{F}_i$ -measurable.

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In Lean, it is formalized as (this is the version used in the paper):

```
def IsStoppingTime [Preorder  $\iota$ ] ( $\mathcal{F}$  : Filtration  $\iota$   $\mathfrak{F}$ ) ( $\tau$  :  $\Omega \rightarrow \iota$ )  
  :=  $\forall i : \iota$ , MeasurableSet[ $\mathcal{F}$  i] { $\omega$  |  $\tau \omega \leq i$ }
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```
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  :=  $\forall i : \iota$ , MeasurableSet[ $\mathcal{F}$  i] { $\omega$  |  $\tau \omega \leq i$ }
```

Note: In current version of Mathlib, Lean uses $\tau : \Omega \rightarrow \text{WithTop } \iota$ rather than $\tau : \Omega \rightarrow \iota$ to allow $\tau = \infty$ (i.e. the process never stops). We'll talk about this in the end of the talk.

Stopped process

We can define a process that will stop when a particular event occur:

Definition

Let (f_m) be a stochastic process and τ be a stopping time. Then the **stopped process** is defined as $m \in \mathbb{N} \mapsto f_{\tau \wedge m}$.

```
noncomputable def stoppedProcess
  (f :  $\iota \rightarrow \Omega \rightarrow E$ ) ( $\tau : \Omega \rightarrow \iota$ ) :  $\iota \rightarrow \Omega \rightarrow E :=$ 
  fun i  $\omega \Rightarrow$  f (min i ( $\tau$   $\omega$ ))  $\omega$ 
```

The stopped process will evolve as f_m if the event does not occur at time m , and stop as f_τ if the event does occur before time m .

Here's an important example of 'stopping time', which is called **hitting time**:

Example

Given a E -valued process $(f_n)_{n \in \mathbb{N}}$ and set $\mathfrak{F}_n = \sigma(f_i : i \leq n)$. Given a set $A \subset E$, the **hitting time** is thought of as the first time (f_n) enters A . Namely, it's defined to be:

$$H^f(A) : \Omega \rightarrow \mathbb{N} \cup \{\infty\}, \quad \omega \mapsto \inf\{n \mid f_n(\omega) \in A\}$$

and we define $\inf \emptyset = \infty$. We can observe that if $\{n \mid f_n(\omega) \in A\} \neq \emptyset$, then

1. $f_{H^f(A)} \in A$;
2. $\forall n < H^f(A), f_n \notin A$.

Similar as stopped process, we define a 'hitting process' by $m \in \mathbb{N} \mapsto f_{H^f(A) \wedge m}$ where $m \wedge \infty = m$.

However, (f_n) is indexed by $\mathbb{N}$, while $H^f(A)$ takes value in $\mathbb{N} \cup \{\infty\}$, so $f_{H^f(A)}$ is not allowed in a type system like Lean.

The way to fix this issue is to incorporate the stopped process notion into our definition of hitting time, i.e. we define something like ' $H^f(A) \wedge m$ ' as our hitting time rather than $H^f(A)$:

- ▶ By introducing an ending time m , we define the hitting time to be the first time (f_n) enters A before time m ;
- ▶ Moreover, in order to also work for the case that the time index is unbounded from below, it will be useful to introduce a starting time n , which results the following definition:

Definition

Given a process $(f_i)_{i \in \iota}$, a set $A \subseteq E$, and times $n \leq m$ in ι , the **hitting time** $H_{n,m}^f(A) : \Omega \rightarrow \iota$ between n and m is defined by:

$$H_{n,m}^f(A)(\omega) = \begin{cases} \inf\{i \in [n, m] \mid f_i(\omega) \in A\} & \text{if } \exists j \in [n, m], f_j(\omega) \in A \\ m & \text{otherwise} \end{cases}$$

In Lean, this is formalized as:

```
noncomputable def hittingBtwn (f :  $\iota \rightarrow \Omega \rightarrow E$ )  
  (s : Set E) (n m :  $\iota$ ) :  $\Omega \rightarrow \iota$  :=  
  fun  $\omega \Rightarrow$  if  $\exists j \in \text{Set.Icc } n \ m, f \ j \ \omega \in s$   
    then sInf (Set.Icc n m  $\cap \{i : \iota \mid f \ i \ \omega \in s\}$ ) else m
```

Moreover, our new definition of hitting time is also a stopping time:

```
theorem Adapted.isStoppingTime_hittingBtwn  
  [ConditionallyCompleteLinearOrder  $\iota$ ] [WellFoundedLT  $\iota$ ] [Countable  $\iota$ ]  
  { _ : MeasurableSpace E} { $\mathcal{F}$  : Filtration  $\iota$   $\mathfrak{F}$ }  
  {f :  $\iota \rightarrow \Omega \rightarrow E$ } {s : Set E} {n n' :  $\iota$ }  
  (hf : Adapted  $\mathcal{F}$  f) (hs : MeasurableSet s) :  
  IsStoppingTime  $\mathcal{F}$  (fun  $\omega \Rightarrow$  hittingBtwn f s n n'  $\omega$ )
```

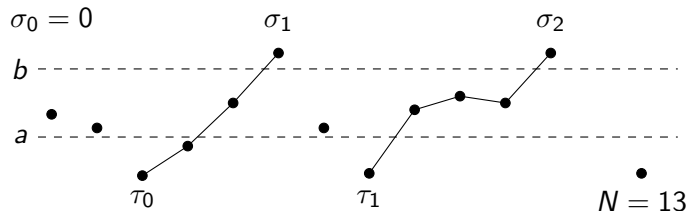
Upcrossing

Given real values $a < b$, we would like to define a notion called **upcrossing** of f_n across the band $[a, b]$ which counts the number of time (f_n) crosses from below a to above b . To make it rigorous, we define this number as follow: first, we define

$$\sigma_0 := 0, \tau_n := H_{\sigma_n, N}^f(-\infty, a], \sigma_{n+1} := H_{\tau_{n+1}, N}^f[b, \infty),$$

for some given $N \in \mathbb{N}$. Then, we define the upcrossings of (f_n) before time N to be:

$$U_N^f(a, b) := \sup\{n | \sigma_n < N\}.$$



Here's the formalization of upcrossing in Lean:

```
noncomputable def lowerCrossingTimeAux [InfSet  $\iota$ ]  
  (a :  $\mathbb{R}$ ) (f :  $\iota \rightarrow \Omega \rightarrow \mathbb{R}$ ) (c N :  $\iota$ ) :  $\Omega \rightarrow \iota :=$   
  hittingBtwn f (Set.Iic a) c N
```

```
noncomputable def upperCrossingTime [OrderBot  $\iota$ ] [InfSet  $\iota$ ]  
  (a b :  $\mathbb{R}$ ) (f :  $\iota \rightarrow \Omega \rightarrow \mathbb{R}$ ) (N :  $\iota$ ) :  $\mathbb{N} \rightarrow \Omega \rightarrow \iota$   
  | 0 =>  $\perp$   
  | n + 1 => fun  $\omega$  =>  
    hittingBtwn f (Set.Ici b)  
    (lowerCrossingTimeAux a f (upperCrossingTime a b f N n  $\omega$ ) N  $\omega$ ) N  $\omega$ 
```

```
noncomputable def upcrossingsBefore [OrderBot  $\iota$ ] [InfSet  $\iota$ ]  
  (a b :  $\mathbb{R}$ ) (f :  $\iota \rightarrow \Omega \rightarrow \mathbb{R}$ ) (N :  $\iota$ ) ( $\omega$  :  $\Omega$ ) :  $\mathbb{N} :=$   
  sSup {n | upperCrossingTime a b f N n  $\omega$  < N}
```

Doob's upcrossing estimate

Recall the stock price model of (sub/super)-martingale. Now let (f_n) be the stock price of a declining business, which can be viewed as supermartingale. One might naively think "buy low sell high" will be a plausible 'winning' strategy. We can formulate this idea into a predictable process as follow:

$$c_1 := 1_{f_0 < a}, \quad c_n := 1_{c_{n-1}=1} 1_{f_{n-1} \leq b} + 1_{c_{n-1}=0} 1_{f_{n-1} < a} \text{ for } n \geq 2.$$

Each time there is an upcrossing, one will win at least $b - a$, thus at time N ,

$$(c \cdot f)_N \geq (b - a)U_N^f(a, b) - (a - f_N)^+$$

where g^+ denotes the positive part of the function g .

However, recall the 'no-system theorem', $(c \cdot f)$ is a supermartingale, so we have $0 = \mu[(c \cdot f)_0] \geq \mu[(c \cdot f)_N]$, therefore we get:

$$(b - a)\mu[U_N^f(a, b)] \leq \mu[(a - f_N)^+]$$

If we denote $U^f(a, b) := \sup_{N \in \mathbb{N}} U_N^f(a, b)$, then we have **Doob's upcrossing estimate** for supermartingale:

$$(b - a)\mu[U^f(a, b)] \leq \sup_{N \in \mathbb{N}} \mu[(a - f_N)^+].$$

Similarly, we have Doob's upcrossing estimate for submartingale as well:

$$(b - a)\mu[U_N^f(a, b)] \leq \mu[(f_N - a)^+], \quad (b - a)\mu[U^f(a, b)] \leq \sup_{N \in \mathbb{N}} \mu[(f_N - a)^+].$$

We can view (f_n) as the stock price of a growing business. One might think "buy low sell high" strategy as a 'better' strategy than simply "buy and hold", as it makes benefits for every fluctuation, but this estimate in some sense gives a counter-intuition.

This estimate will be useful when we prove martingale convergence theorem later.

Martingale convergence theorems

Let $(\Omega, \mathfrak{F}, \mu, (\mathfrak{F}_n)_{n \in \mathbb{N}})$ be a filtered space. Given a martingale $(f_n)_{n \in \mathbb{N}}$ on Ω , it is natural to ask: under what condition, (f_n) has a limit function f as $n \rightarrow \infty$. The following theorem tells us that L^1 -boundedness gives a.e. convergence to that limit:

Theorem (Almost everywhere martingale convergence)

Given a (sub)martingale $(f_n)_{n \in \mathbb{N}}$ adapted to filtration $(\mathfrak{F}_n)_{n \in \mathbb{N}}$, if there exists some $R \in \mathbb{R}$ s.t. $\|f_n\|_{L^1} := \int |f_n| \leq R$ for all n , then there exists some $\mathfrak{F}_\infty := \bigvee_{n \in \mathbb{N}} \mathfrak{F}_n$ -measurable function f which is integrable and $f_n \xrightarrow{\text{a.e.}} f$.

The next result characterizes the L^1 -convergence of martingales:

Theorem (L^1 martingale convergence)

Given a martingale $(f_n)_{n \in \mathbb{N}}$ adapted to filtration $(\mathfrak{F}_n)_{n \in \mathbb{N}}$, the following are equivalent:

- ▶ *(f_n) is uniformly integrable (in probability sense);*
- ▶ *(f_n) converges in L^1 to some $\mathfrak{F}_\infty$ -measurable, integrable function f ;*
- ▶ *there exists some $\mathfrak{F}_\infty$ -measurable and integrable f s.t. for all $n \in \mathbb{N}$, $\mu[f|\mathfrak{F}_n] = f_n$ a.e..*

We will discuss uniform integrability later. These two theorems are the core of this paper, which are very important in the study of martingale. They can for example be used to prove Kolmogorov's 0-1 law, the strong law of large numbers, etc.

Uniform Integrability

In the literature, there are two non-equivalent formulations of uniform integrability used by analysts and probabilists resp.:

Definition (Uniform integrability, analyst)

A sequence of functions $(f_n) \subset L^1(E; \mu)$ is said to be **uniformly integrable** if for all $\varepsilon > 0$, there exists some $\delta > 0$ s.t. for all $A \in \mathfrak{F}$ with $\mu(A) < \delta$, we have $\|f_n 1_A\|_{L^1} < \varepsilon$ for all n .

Definition (Uniform integrability, probabilist)

A sequence of functions $(f_n) \subset L^1(E; \mu)$ is said to be **uniformly integrable** if

$$\lim_{C \rightarrow \infty} \sup_n \|f_n 1_{\{|f_n| \geq C\}}\|_{L^1} = 0$$

Uniform integrability in probabilist's sense is strictly stronger than the analyst's sense. These two definitions are linked by following theorem:

Theorem

If μ is a finite measure, then $(f_n) \subset L^1(E, \mu)$ is uniformly integrable in probabilist's sense if and only if it is uniformly bounded in L^1 and is uniformly integrable in the analyst's sense.

In Lean, these two definitions are formulated as:

```
def UnifIntegrable (f :  $\iota \rightarrow \Omega \rightarrow E$ ) (p :  $\mathbb{R}_{\geq 0\infty}$ ) ( $\mu$  : Measure  $\Omega$ ) : Prop :=
   $\forall \{\varepsilon : \mathbb{R}\} (\_ : 0 < \varepsilon), \exists (\delta : \mathbb{R}) (\_ : 0 < \delta), \forall i \ s,$ 
    MeasurableSet s  $\rightarrow \mu \ s \leq \text{ENNReal.ofReal } \delta \rightarrow$ 
    eLpNorm (s.indicator (f i)) p  $\mu \leq \text{ENNReal.ofReal } \varepsilon$ 
```

```
def UniformIntegrable (f :  $\iota \rightarrow \Omega \rightarrow E$ ) (p :  $\mathbb{R}_{\geq 0\infty}$ ) ( $\mu$  : Measure  $\Omega$ ) : Prop
:= ( $\forall i, \text{AEStronglyMeasurable (f i) } \mu$ )  $\wedge$  UnifIntegrable f p  $\mu \wedge$ 
   $\exists C : \mathbb{R}_{\geq 0}, \forall i, \text{eLpNorm (f i) p } \mu \leq C$ 
```

We note that both definitions has an extra parameter $p : \mathbb{R}_{\geq 0\infty}$, and our original definition can be recovered if $p = 1$.

Almost everywhere (sub)martingale convergence theorem

Theorem

Given a submartingale $(f_n)_{n \in \mathbb{N}}$ adapted to filtration $(\mathfrak{F}_n)_{n \in \mathbb{N}}$, if there exists some $R \in \mathbb{R}$ s.t. $\|f_n\|_1 \leq R$ for all n , then there exists some $\mathfrak{F}_\infty$ -measurable function f which is integrable and $f_n \xrightarrow{a.e.} f$.

```
theorem Submartingale.ae_tendsto_limitProcess {F : Filtration ℕ ℱ}
  {f : ℕ → Ω → ℝ} {R : ℝ≥0} [IsFiniteMeasure μ]
  (hf : Submartingale f F μ) (hbdd : ∀ n, eLpNorm (f n) 1 μ ≤ R) :
  ∀m ω ∂μ, Tendsto (fun n => f n ω) atTop (ℕ (F.limitProcess f μ ω))
```

```
noncomputable def limitProcess (f : ι → Ω → E) (F : Filtration ι ℱ)
  (μ : Measure Ω) :=
if h : ∃ g : Ω → E,
  StronglyMeasurable[ ⋔ n, F n] g ∧
  ∀m ω ∂μ, Tendsto (fun n => f n ω) atTop (ℕ (g ω)) then
  Classical.choose h else 0
```

So in Lean, instead stating the existence of a limiting random variable, it defines the limiting random variable if it does exist using axiom of choice. The benefits from doing so are as follow:

- ▶ We avoid needing to deconstruct the a.e. martingale convergence theorem every time when we need to manipulate the limiting random variable;
- ▶ We can reason about the definition itself, e.g.

```
theorem stronglyMeasurable_limitProcess :  
  StronglyMeasurable[ $\bigsqcup$  n,  $\mathcal{F}$  n] (limitProcess f  $\mathcal{F}$   $\mu$ )
```

```
theorem memLp_limitProcess_of_eLpNorm_bdd {R :  $\mathbb{R}_{\geq 0}$ } {p :  $\mathbb{R}_{\geq 0\infty}$ }  
  {F : Type*} [NormedAddCommGroup F]  
  { $\mathcal{F}$  : Filtration  $\mathbb{N}$   $\mathfrak{F}$ } {f :  $\mathbb{N} \rightarrow \Omega \rightarrow F$ }  
  (hfm :  $\forall$  n, AESTronglyMeasurable (f n)  $\mu$ )  
  (hbdd :  $\forall$  n, eLpNorm (f n) p  $\mu \leq R$ ) :  
  MemLp (limitProcess f  $\mathcal{F}$   $\mu$ ) p  $\mu$ 
```


Proof of a.e. martingale convergence theorem: sketch

Note that:

Lemma

A real sequence (x_n) converges if and only if $(|x_n|)$ is bounded above and for all $a < b \in \mathbb{Q}$, (x_n) does not visit below a and above b infinitely often.

So to prove the convergence, it suffices to show that

- ▶ $(|f_n|)$ is bounded a.e.;
- ▶ (f_n) does not visit below a and above b infinitely often a.e..

As (f_n) is L^1 -bounded by R , by Doob's upcrossing estimate, we have:

$$(b - a)\mu[U^f(a, b)] \leq \sup_{n \in \mathbb{N}} \mu[(f_n - a)^+] \leq R + |a| \cdot \mu(\Omega) < \infty$$

Then as $U^f(a, b) < \infty$ a.e., (f_n) does not visit below a and above b infinitely often a.e..

For a.e.-boundedness, it's a technical analysis proof so I skip here. To prove this a.e. limit function is integrable and $\mathfrak{F}_\infty$ -measurable, I also skip here.

Application: Lévy's generalized Borel-Cantelli lemma

One of the most important applications of martingale convergence theorem is **Lévy's generalized Borel-Cantelli lemma**:

Theorem

Given a filtration $(\mathfrak{F}_n)_{n \in \mathbb{N}}$ and a sequence of sets $(S_n)_{n \in \mathbb{N}}$ s.t. $S_n \in \mathfrak{F}_n$ for all n , then

$$\limsup_{n \rightarrow \infty} S_n = \left\{ \sum_{n=0}^{\infty} \mu[1_{S_{n+1}} | \mathfrak{F}_n] = \infty \right\}$$

almost everywhere.

We will omit the proof here.

But one of the issue the paper encounter is: in order to prove above theorem, it needs to show that given f_n : submartingale, consider a hitting time:

$$\sigma := \inf\{n | f_n \geq r\} = H^f[r, \infty)$$

the hitting process $f_{\sigma \wedge n}$ is a submartingale. It is a standard result for stopped process:

Theorem

If f_n is a submartingale and τ is a stopping time. The stopped process $f_{\tau \wedge n}$ is then a submartingale.

However, recall at the moment, our stopping time cannot take value to be $\tau = \infty$, while σ can take value to be ∞ .

So the solution in this paper is to give up on using this general result of stopped process, and define:

$$f_{\sigma \wedge n} := f_{H_{[0,n]}^f}[r, \infty)$$

but still, we cannot write this hitting process in the form of stopping process. So the author has to prove this hitting process satisfies certain properties that shares with stopped process, which causes lots of code duplication.

Stopping time: revisit

So in current version of Mathlib, Lean uses $\tau : \Omega \rightarrow \text{WithTop } \iota$ rather than $\tau : \Omega \rightarrow \iota$ to allow $\tau = \infty$:

```
def IsStoppingTime [Preorder  $\iota$ ] ( $\mathcal{F}$  : Filtration  $\iota$   $\mathfrak{F}$ ) ( $\tau$  :  $\Omega \rightarrow \text{WithTop } \iota$ )  
  :=  $\forall i : \iota$ , MeasurableSet[ $\mathcal{F}$  i] { $\omega \mid \tau \omega \leq i$ }
```

and revise the definition of stopped process as follow:

```
def stoppedProcess (f :  $\iota \rightarrow \Omega \rightarrow E$ ) ( $\tau$  :  $\Omega \rightarrow \text{WithTop } \iota$ ) :  $\iota \rightarrow \Omega \rightarrow E$  :=  
  fun i  $\omega$  => f (min (i : WithTop  $\iota$ ) ( $\tau \omega$ )).untopA  $\omega$ 
```

Then our definition of hitting time, which takes value of ∞ , will be then an example of stopping time:

```
noncomputable def hittingAfter (f :  $\iota \rightarrow \Omega \rightarrow E$ ) (s : Set E) (n :  $\iota$ ) :
   $\Omega \rightarrow \text{WithTop } \iota :=$ 
  fun x  $\mapsto$  if  $\exists j, n \leq j \wedge f j x \in s$  then (sInf {i :  $\iota$  |  $n \leq i \wedge f i x \in s$ 
    } :  $\iota$ ) else  $\top$ 
```

and our $H^f(A) = \inf\{n | f_n \in A\}$ will be written as `hittingAfter f A  $\perp$` .

```
theorem Adapted.isStoppingTime_hittingAfter
  [ConditionallyCompleteLinearOrder  $\iota$ ]
  [WellFoundedLT  $\iota$ ] [Countable  $\iota$ ] {_ : MeasurableSpace E} { $\mathcal{F}$  : Filtration
     $\iota$   $\mathfrak{F}$ }
  {f :  $\iota \rightarrow \Omega \rightarrow E$ } {s : Set E} {n :  $\iota$ } (hf : Adapted  $\mathcal{F}$  f)
  (hs : MeasurableSet s) : IsStoppingTime  $\mathcal{F}$  (hittingAfter f s n)
```

Then we can use all the theory developed in stopped process to our hitting process.

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```

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   $\Omega \rightarrow \text{WithTop } \iota :=$ 
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    } :  $\iota$ ) else  $\top$ 

```

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```

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  [WellFoundedLT  $\iota$ ] [Countable  $\iota$ ] { _ : MeasurableSpace E} { $\mathcal{F}$  : Filtration
     $\iota$   $\mathfrak{F}$ }
  {f :  $\iota \rightarrow \Omega \rightarrow E$ } {s : Set E} {n :  $\iota$ } (hf : Adapted  $\mathcal{F}$  f)
  (hs : MeasurableSet s) : IsStoppingTime  $\mathcal{F}$  (hittingAfter f s n)

```

Then we can use all the theory developed in stopped process to our hitting process. But as the cost, one can not index a process directly with stopping time like f_τ .

Questions?